the posterior probabilities of models $Pr(M_1|\boldsymbol{y})$ and $Pr(M_2|\boldsymbol{y})$ proves computationally prohibitive that reliance on a Bayes factor is clearly justified. We also find that the careful assessment of models, through the use of posterior predictive checks on model behavior can be a more informative way to compare models than just a one number representation of posterior odds ratio (Bayes factor) or even the actual posterior odds. This is because the intelligent selection of test quantities in a posterior predictive assessment can provide more detailed information about the way in which a model might fail to provide a good representation of structure in the data.

19.2.3 Testing Goodness of Fit

In Chapter 17.10.7 we went into some detail regarding the position of Karl Pearson on interpretation of $p-$vlaues associated with his Chi-squared GOF test and, by extension, other GOF tests based on measures of discrepancy between what is expected under a hypothesized model and what is observed in the data. Why are the opinions of K. Pearson regarding the interpretation of his GOF test relevant for us? First, because it is clear that he never intended the GOF test to be assessed in terms of $p-$values following either of the routes offered by Fisher's tests of significance or Neyman-Pearson tests for acceptance sampling. Perhaps, then, neither should we. Secondly, K. Pearson's interpretation of the results of a GOF test ask us to reconsider our own views about how such tests fit into the overall framework of testing statistical hypotheses. Are either of the Fisherian or Neyman-Pearson approaches to interpretation of $p-$values truly appropriate for GOF tests? If not, then how should the results of GOF tests be interpreted?

We see two avenues for the interpretation of $p-$values calculated from

GOF tests. One is the Fisherian interpretation of a $p-$value as a measure of evidence against the hypothesis, which here is a theoretical probability distribution. Under this interpretation we have the usual conceptualization of looking at the distribution of a test statistic under the assumption that the hypothesized distribution is correct, and determining whether the actual test statistic has a value that would be unusual under that distribution, or not. Unusual generally means the use of tail areas which, as we have discussed previously, is not without some controversy. The controversy over tail areas aside, this conceptualization basically considers the observed data to be a realization from the posited model, and hence Fisher's explanation that a $p-$value represents the probability of obtaining a result as extreme or more extreme than that actually observed if the hypothesis is true. If we choose this route for interpretation of GOF tests we are left with the conundrum of whether a $p-$value of 0.07, say, offers sufficiently little evidence against the hypothesized distribution that we fail to reject it, or whether it offers sufficiently strong evidence against the hypothesized distribution to declare it untenable as a description of the observed data. Perhaps an even more serious disconnect between GOF tests and the Fisherian significance testing framework is that in the Fisherian framework, a hypothesis that we fail to reject is simply ignored whereas, in a GOF test, we wish to declare positive support for a hypothesized model. The typical approach among applied statisticians is to simply ignore our conundrum, since we have no good solution to it, and hope for an unambiguous $p-$value that is either small (e.g., less than 0.05) or decidedly not small (e.g., $p > 0.15$).

The other possible framework under which to interpret $p-$values in GOF tests is essentially that of Pearson. Consider, for simplicity, a one sample problem in which our hypothesized model is that $Y_1, \ldots, Y_n$ are indepen-

dent and identically distributed random variables with distribution function $G(y|\boldsymbol{\theta})$. We have data, $y_1, \ldots, y_n$, which may or may not correspond to the hypothesized model, including the stipulations of independence and identical distribution as well as the distributional form. Now, if the observed data are a random sample from $G(y|\theta)$, then the empirical distribution $F_n(x)$ converges to $G(x|\theta)$. But even if this is not true, and the data are not a random sample from $G(y|\theta)$, it is possible that $G(x|\theta)$ provides a good description of the frequency distribution $F_n(x)$. Our objective, then, is to determine whether the hypothesized model $G(y|\theta)$ provides a good representation of the data, regardless of where the data actually came from. The important point is that we are concerned with testing the degree to which a theoretical model describes the frequency distribution of our data, not whether the data actually arose from the theoretical model. Our test statistic is in the form of a discrepancy measure, such as the χ^2 statistic, deviance, or a Kolmogorov statistic. These quantities all have the propertie s of being non-negative and monotone increasing as the degree of discrepancy gets larger. The $p-$value for a perfect fit (model frequencies equal to data frequencies) is $p = 1.0$. There is no concern for tail areas as representing regions of low probability, only the fact that as the test statistic increases from zero, the $p-$value decreases from 1.0. The value of p is that it functions on the same scale for all suitable problems, while the absolute values of test statistics can be quite different among problems. There is no definitive threshold, "The value of P at which we consider goodness or badness of graduation starts cannot be fixed without regard to the special problem under consideration" (Pearson, 1935). But given a monotone relation between "goodness of graduation" and $p-$values, we can declare relative differences among competing models. This is, although a bit subtle, actually quite a departure from the usual interpre-

tation of $p-$values in tests of significance and not even in the same universe with the interpretation in acceptance sampling in which different $p-$values carry no meaning except as to which side of the critical value they are on.

Simulation-based GOF procedures, such as those discussed in Chapter 11 or those based on posterior predictive $p-$values in a Bayesian analysis can be interpreted by considering $p-$values in much the same manner as for tests of significance. Although this does not solve the problem of whether or not to declare an adequate fit based on a $p-$value of small but not extremely small magnitude (e.g., $0.05 < p < 0.15$), simulation based methods lend themselves to multiple assessments using different characteristics of data structure. For example, in a regression problem with multiple observations at each covariate value, we might construct one test statistic as the average difference between the maximum data value and the third quartile at each level of the covariate to reflect extreme tail behavior. We might then construct another test statistic as the slope in a Box-Cox plot to reflect mean-variance relation, and yet another test statistic as the value of a Kolmogorov statistic to compare generalized residuals to a uniform distribution to reflect distributional form. This would then result in three simulation-based $p-$values of the type discussed in Chapter 11. Far from being the hindrance we are taught about multiple testing (from the viewpoint of Neyman-Pearson theory) having three $p-$values allows us to draw more detailed conclusions about which aspects of data behavior are well represented by the model and about which aspects of data behavior might be not so well represented by the model. It is normal, not unusual, that for any reasonably complex situation, models will describe different parts of data behavior with differential quality. Thus, we may have several $p-$values that vary in magnitude because they are measuring different characteristics of the model. The same is true of tests developed from

posterior predictive distributions in a Bayeisn analysis. In fact, we find the force of this concept so compelling that we are ready to recommend that statisticians abandon the search for one omnibus GOF procedure that will declare a model *adequate* or *inadequate* as a whole, except in the simplest of problems.

19.3 Overall Inferential Frameworks

As we have indicated several times, we do not believe there exists a single approach to statistical analysis that should be followed in every problem. This does not mean, however, that we find any approach equally appropriate regardless of the problem. And, there are methodological areas we have not touched upon, such as functional data analysis, random forests, and nonparametric smoothers, just to name a few. Whether there are definitive philosophical underpinnings for these less traditional methods that can be compared and contrasted with those we have presented here can be considered an open question. It has been our position throughout the book that if the only objective of an analysis is pure data description or prediction, then one is free of concerns connected with the representation of scientific mechanisms by stochastic models. Thus, while not dismissing procedures that focus solely on data description or prediction as lacking value, we do find that their objectives lie somewhat outside of the sphere of statistical inference as it relates to understanding and explanation. In this section, then, we briefly discuss the types of considerations that are relevant in determining the inferential framework one will apply to a given problem.